Exercise Sheet 7

Structural Balance

5942UE Network Science

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7.A Triadic Balance Theory

Exercise 7.A.1: Classifying Signed Triangles

Label each signed triangle as **balanced (B)** or **unbalanced (U)**:

1. $A \overset{+}{-} B, B \overset{+}{-} C, A \overset{+}{-} C$ (all positive)
2. $A \overset{+}{-} B, B \overset{+}{-} C, A \overset{-}{-} C$ (two positive, one negative)
3. $A \overset{-}{-} B, B \overset{-}{-} C, A \overset{+}{-} C$ (one positive, two negative)
4. $A \overset{-}{-} B, B \overset{-}{-} C, A \overset{-}{-} C$ (all negative)

Which pattern is most stable, and which is "socially unstable"?

Exercise 7.A.2: Applying the Balance Theorem

Complete signed graph on $1, 2, 3, 4, 5$. All edges positive **except** 1–3, 1–4, 2–3, 2–4, 5–3, and 5–4 (which are negative).

1. Check whether every triangle is balanced.
2. If balanced, find the **two-camp partition** guaranteed by the Balance Theorem.
3. Verify within-camp edges are positive and between-camp edges negative.
4. What does the partition represent socially?

7.B Weak Balance

Exercise 7.B.1: Strong vs. Weak Balance

Graph with two triangles connected by one edge:

- A, B, C all positive
- D, E, F all negative
- Connecting edge C–D is negative

1. Is A, B, C balanced under strong balance? Under weak balance?
2. Is D, E, F balanced under strong balance? Under weak balance?
3. What does the global partition look like under each?
4. In international relations, what does an all-negative triangle represent?

Exercise 7.B.2: Finding Camps in a Signed Network

A signed graph: $1, 2, 3$ are mutually friends, $4, 5, 6$ are mutually friends, all cross-group edges are negative.

1. Check every triangle for balance.
2. Find the two-camp partition.
3. Verify within-group edges are positive, between-group negative.
4. Flip the cross-group edge 1–4 from negative to positive. Which triangles become unbalanced?

7.C Relaxed Balance and Applications

Exercise 7.C.1: WWI Alliance Network

Five countries during WWI:

- **Allies**: France, Britain, Russia (mutually positive)
- **Central Powers**: Germany, Austria-Hungary (mutually positive)
- All Allies–Central Powers edges are negative

1. Is this network perfectly balanced? Identify any unbalanced triangles.
2. Does it match the two-camp Balance Theorem?
3. What if a new country joins with positive ties to **both** camps?
4. Italy switched from Germany to the Allies in 1915 — what does balance theory predict?

Exercise 7.C.2: Approximate Balance in Real Networks

Using the karate club graph, sign edges by faction (within = +, cross = −), then flip a small number of cross-faction edges to positive to simulate noisy or ambiguous relationships.

1. Count balanced vs. unbalanced triangles.
2. What fraction of triangles are balanced?
3. Remove edges in order of how many unbalanced triangles they participate in. After removing the top 5, how does the fraction change?
4. Plot fraction-balanced vs. edges-removed.

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Wrap-Up

- triangle sign products give a one-line balance test
- the Balance Theorem turns a local rule into a global two-camp partition
- weak balance permits more than two camps and is often more realistic
- real networks are rarely perfectly balanced — approximate balance is the typical regime